

Jimmy Gu

I take messy real-world data from *raw capture* to a *tested result*.

I'm a physics (biophysics) student at WashU, graduating in December 2026. My projects start with instrument sweeps, donation exports, stock prices and image sets, and end with a result checked against a baseline: coil theory, a published study, the S&P 500. I report what didn't hold up, too.

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EDUCATION

B.A. Physics (Biophysics), Biology minor

Washington University in St. Louis · GPA 3.83 · Dec 2026

LEADERSHIP

Cadet, U.S. Air Force Academy

Superintendent's List · coordinated mission planning for 100+ cadets

FOUNDER

Journey of Hope nonprofit

Ran it for 10+ years · \$15,000+ raised · up to 20 volunteers

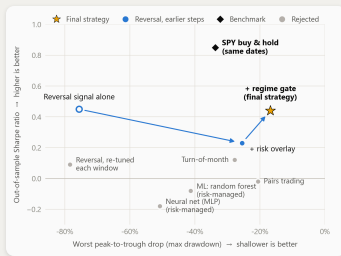
TOOLS

Python and MATLAB

pandas · SciPy · scikit-learn · PyTorch · Streamlit · Git

Four projects, one page each

Each page leads with the outcome and objective, then my role, annotated figures, results and what I learned. Select a project to jump to its page.



The main strategies tested, scored out of sample. Gray points were rejected.

QUANT RESEARCH · SOLO · 2026

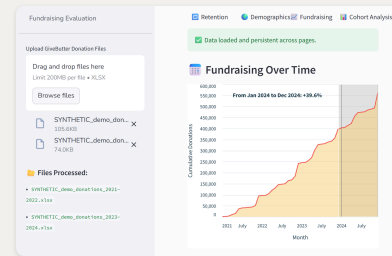
PAGE 2 →

Stress-testing trading strategies until only one survived

0.44 -16.8%

Sharpe ratio (return per unit of risk), out of Worst drop, half the S&P 500's sample

Led an out-of-sample study of trading strategies on 10 years of prices: one survived, but it earned less than the S&P 500.



The dashboard's Home page, running on **synthetic demo data**, not the client's.

CLIENT ANALYTICS · CONSULT YOUR COMMUNITY · 2025

PAGE 3 →

A donor dashboard that shows a crime victim nonprofit which donors come back

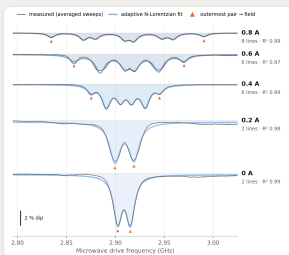
6

Analyses from one upload: trend, retention, cohorts, top donors, map, campaigns

3 yrs

Of campaign totals compared in my first-pass analysis (2022 to 2024)

Analyzed a crime victim nonprofit's donor records, then built a Streamlit dashboard that repeats the analysis on each new GiveButter export.



Real averaged spectra (gray) and my automatic fits (blue) as the field grows.

INSTRUMENTATION · TEAM OF 2 · 2026

PAGE 4 →

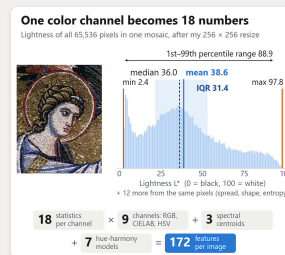
Turning 10,000 noisy sweeps into a working diamond magnetic-field sensor

10,000 $R^2 \geq 0.97$

Sweeps captured, 85% kept after automated QC

Fit quality in all 10 field conditions

Built the Python pipeline behind a diamond magnetic-field sensor: it drives the instruments, cleans the sweeps and fits every spectrum.



What my extractor computes for one channel. Art: P. Cavallini (public domain).

NEUROAESTHETICS RESEARCH · WASHU · 2025

PAGE 5 →

Measuring why people like mosaic art: a benchmarked 982-image feature pipeline

982

Mosaic images turned into a feature table

172

Features per image: color, brightness, detail

Wrote the MATLAB feature extractor for a WashU mosaic-art study; it reproduces a published study's average color and brightness measures.

Stress-testing trading strategies until only one survived

I led a study of trading ideas on 10 years of real stock prices. The survivor, a risk-controlled reversal strategy, earned a **Sharpe ratio of 0.44** (return per unit of risk) with a **worst drop of 16.8%**, half the S&P 500's, but returned less than the index.

OBJECTIVE Test whether a small account can capture a well-documented market pattern too small for large funds, the way a quant researcher would: a literature hypothesis, a realistic backtest, true out-of-sample validation, and honest reporting of what failed.

0.44

Out-of-sample Sharpe, after costs

-16.8%

Max drawdown (S&P 500: -33.7%)

37

Walk-forward test windows, 9+ years

29

Unit tests guarding the pipeline

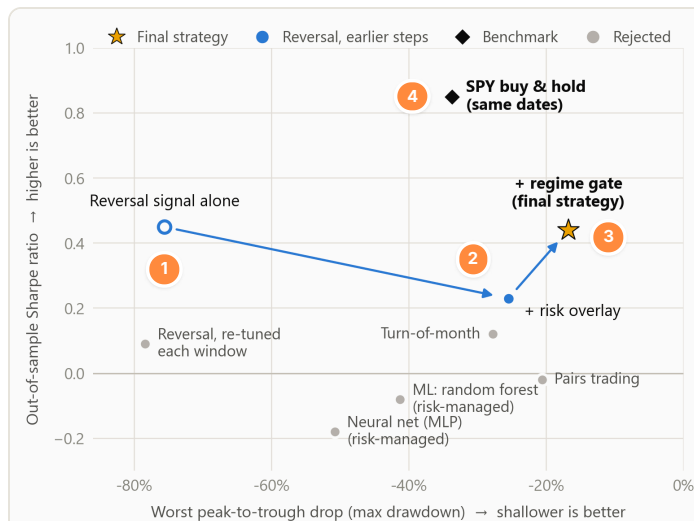


Fig. 1. Every main strategy, scored the same way: out of sample, after trading costs (plotted from the memo's results tables). Gray points were rejected.

- 1 The raw reversal signal (buy recent losers) held up, but its 75.6% worst drop would wipe out most of a small account.
- 2 A risk overlay cut the worst drop to 25.5%, but halved the Sharpe to 0.23.
- 3 A filter that sits out the calmest markets restored a 0.44 Sharpe at a 16.8% worst drop: the final strategy.
- 4 Holding the S&P 500 (SPY) scored higher (0.85), with a deeper 33.7% drop.

MY CONTRIBUTION

Research lead: set the goals and pass bars, signed off on every stage

- **Set the goal:** a strategy a small account could run, tested the way a quant researcher would before risking money.
- **Directed** an AI coding agent (Claude Code) to build the pipeline: cleaning, a realistic backtest, walk-forward tests, a risk overlay.
- **Directed** four follow-up studies, from volatility filters to ML signal searches; a library survey caught a wrong Sortino formula.
- **Set the bar** the ML search had to clear, before it ran.
- **Signed off** on each stage through three pull requests, backed by 29 unit tests.

Python pandas scikit-learn PyTorch statsmodels

Walk-forward validation Permutation testing Unit testing

Data cleaning

AGAINST THE OBJECTIVE

- ✓ **MET** Literature-based idea held up out of sample
- ✓ **MET, WITH LIMITS** Realistic backtest (flat costs, survivorship bias)
- ⚠ **PARTLY** Out-of-sample test: probabilistic Sharpe 0.91, short of 0.95
- ✓ **MET** Honest reporting: every failed idea documented
- ⚠ **NOT MET** Beating the market (SPY Sharpe 0.85)

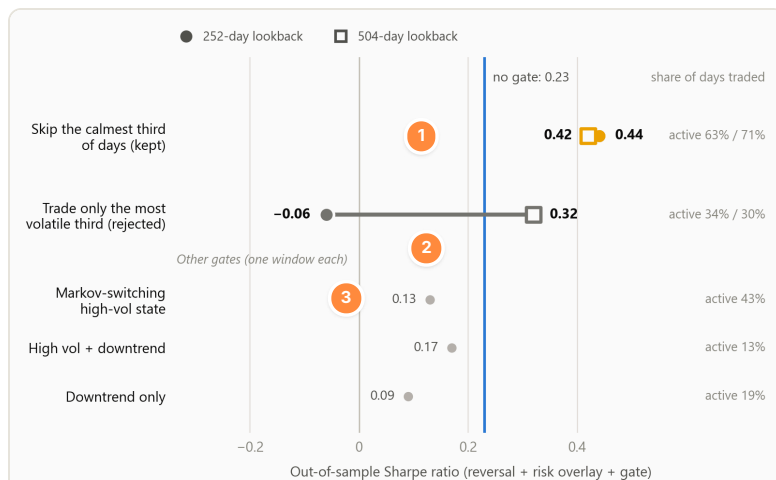


Fig. 2. Process: the robustness check behind the final strategy. The two volatility-tertile filters were each scored at two equally reasonable lookahead windows.

- 1 "Skip the calmest third of days" held up at both (0.44, 0.42): kept.
- 2 "Trade only the most volatile third": 0.32, then -0.06, a fit to one setting: rejected.
- 3 A Markov-switching model scored 0.13, below the no-gate 0.23.

30 candidate stocks, 5 sectors

23 kept · 7 dropped

Regional banks	WAL	ZION	CHA	SAV	FHN
	WBS	ONB	PNFP	UMBF	ABCB
Industrials	WTS	ITT	KAI	ATKR	CIR
Consumer	BOOT	FIVE	WING	CAKE	SFM
Healthcare	OMCL	ICUI	PDCO	ADUS	AMED
Energy	MTDR	SM	PR	CIVI	MGY

A HTTP 404 on every attempt: ticker retired, likely by M&A (5)

B Ticker reassigned: Yahoo returned an unrelated 13-day series (1)

C Too little history: under the 500-day minimum (1)

Clean panel: 23 stocks · 2016-06-10 to 2026-09-18 · 2,583 trading days

Fig. 3. Data cleaning: of 30 candidate stocks, 7 were dropped with the reason logged, not silently filled in (drawn from the memo's data section).

CONSTRAINTS Team: Solo · Timeline: Sep 19 to 20, 2026 · Capital: Sized for a few thousand dollars; no leverage · Data: Free public prices (Yahoo Finance) · Costs: 10 bps (0.10%) per trade, likely optimistic

WHAT I LEARNED One good number is not evidence. A filter that scored 0.32 at one lookahead flipped to -0.06 at another, so I now check results across settings.

A donor dashboard that shows a crime victim nonprofit which donors come back

I analyzed Crime Victim Center (CVC) donor records, found that **a small group of donors gave most of the dollars**, and built a **five-page Streamlit dashboard** that repeats that check, plus retention, map and campaign views, on each new GiveButter export. It **fully covers the brief's donor-ranking goal** and partly covers segmentation, tracking and outreach.

OBJECTIVE CVC's brief asked for a donor strategy aimed at high-value and monthly donors. Five of its goals depend on donor data; I rate each under "Against the objective."

6

Analyses from one upload: trend, retention, cohorts, top donors, map, campaigns

5

Donor-data goals in CVC's brief

3 yrs

Of campaign totals compared in my first-pass analysis (2022 to 2024)

8%

Of CVC's 2022 revenue from donations and fundraisers

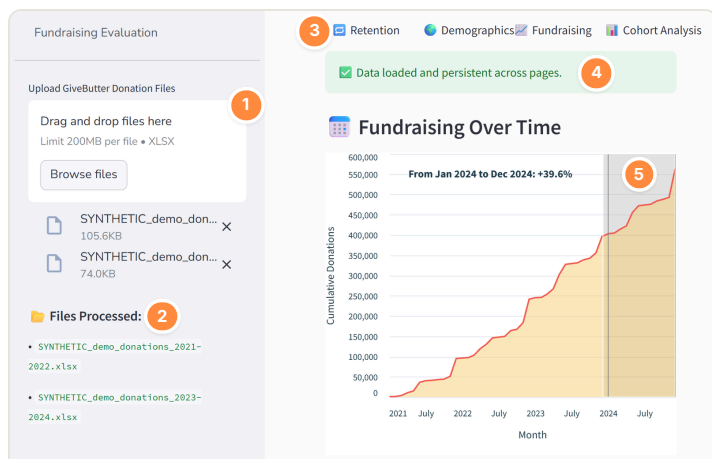


Fig. 1. The Home page after an upload. All screenshots run the repo's code on **synthetic demo data** (1,200 fictional donors), not CVC's, to protect privacy.

- 1 Staff can drop in GiveButter exports, several files at once.
- 2 Every merged file is listed, so staff can see what the numbers include.
- 3 Links to the four analysis pages.
- 4 The cleaned table follows the user to every page for the session.
- 5 Drag across the chart to pick a date range; the app labels the growth over it.

MY CONTRIBUTION

Built the dashboard (sole committer on the public repo); analyzed CVC's donor data

- **Analyzed** CVC's records first: 2022 to 2024 campaign totals and a pandas Pareto (80/20) analysis of donors.
- **Built** the upload-and-clean step that merges any number of GiveButter exports into one table.
- **Designed** the retention metrics: returning vs. one-time donors, quarterly churn, cohort heatmaps.
- **Built** an adjustable top-donor (80/20) view, **geocoded** donor ZIP codes onto a map, and compared campaigns year over year.
- **Wrote** plain-language explainers beside the charts for CVC staff.

Python pandas Streamlit Altair Plotly Data cleaning
Cohort analysis Pareto analysis Geocoding

AGAINST THE OBJECTIVE

- ✓ **MET** Find high-value donors
- ⦿ **PARTLY** Segment donor groups
- ⦿ **PARTLY** Track donors on GiveButter
- ⦿ **INFORMS ONLY** Recurring-gift campaign
- ⦿ **PARTLY** Guide outreach and acquisition
- **NOT MEASURED** Adoption and effect on fundraising

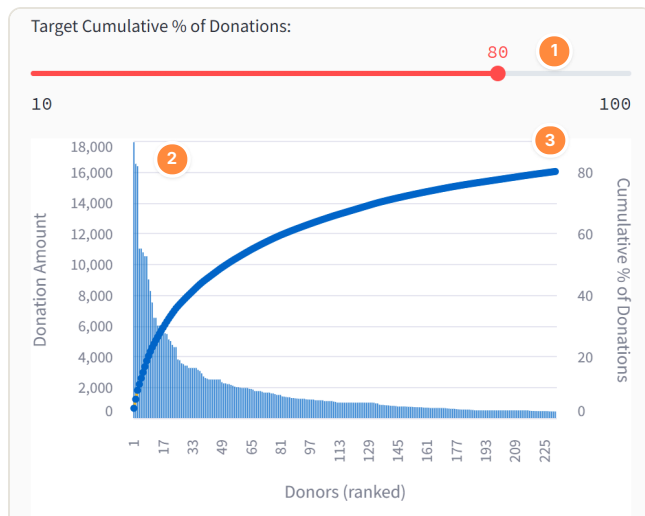


Fig. 2. Top-donor (80/20) view, below a plain-language explainer for staff (synthetic demo data).

- 1 Slider: the share of dollars to explain (default 80%).
- 2 Each donor's total giving in the loaded files, largest first.
- 3 Running share of dollars: 231 of 1,200 demo donors (19%) give 80%.

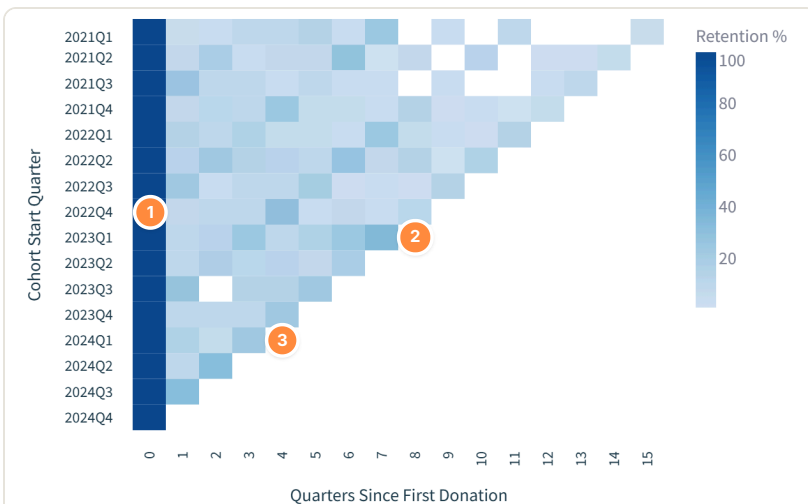


Fig. 3. Which donors come back (synthetic demo data). Rows: first-gift quarter. Color: the share of that cohort giving again in each later quarter.

- 1 Quarter 0 is 100% by definition.
- 2 Read a row left to right to see how a cohort holds up.
- 3 Newer cohorts have had fewer quarters to return: a staircase.

CONSTRAINTS **Team:** WashU student consultants, Consult Your Community · **Timeline:** Spring 2025; code pushed Apr 13 to 14 · **Data:** Private donor records; the repo holds code only · **Input:** GiveButter .xlsx exports in the layout CVC shared

WHAT I LEARNED **Dashboards fail quietly.** A 2026 re-test found four bugs that raise no error (such as a +51,062.7% growth label); I would now ship tests on known data.

Turning 10,000 noisy sweeps into a working diamond magnetic-field sensor

I built the Python pipeline behind a diamond-based magnetic-field sensor. It drives the instruments, captures **10,000 sweeps**, rejects glitched ones and fits every spectrum: up to **8 resonance lines**, $R^2 \geq 0.97$ in all 10 conditions. Readings ran below the coil's predicted field; one constant 0.61 mT offset, fitted afterward and consistent with stray magnetization, brings 0.4 to 0.8 A within about 1.1σ of theory.

OBJECTIVE Build a magnetic-field sensor from teaching-lab hardware and test it against the coil's predicted field. Defects in diamond (NV centers) glow red under a green laser and dim by a few percent at a microwave resonance near 2.9 GHz; a magnetic field splits that resonance, and the split gives the field.

10,000

Sweeps captured (10 currents $\times$ 1,000)

85%

Kept after automated QC (8,517)

 ≥ 0.97 Fit R^2 in every condition 1.1σ

Largest gap to coil theory at 0.4–0.8 A, after one fitted offset

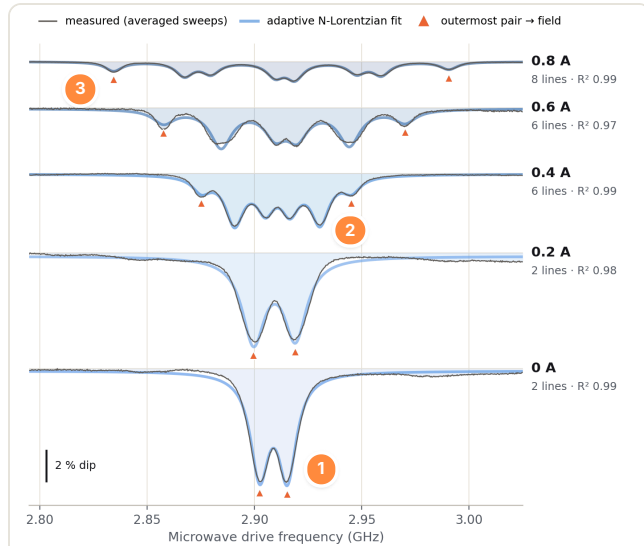


Fig. 1. Real spectra from my pipeline (dark gray) and my adaptive fit (blue) at every other coil current: one resonance splits into many.

- 1 Zero field: a clean double dip, 7.48% deep.
- 2 0.4 A: shallow shoulders a shared-depth fit missed; per-line depths recover them.
- 3 0.8 A: eight lines; the outermost pair (orange) gives the field.

MY CONTRIBUTION

Built all software; sole committer on the repo

- **Automated** an oscilloscope and a function generator from Python (PyVISA/SCPI): 1,000 sweeps per condition.
- **Engineered** QC that rejects glitched sweeps with a data-derived threshold (85% kept).
- **Diagnosed** a trigger that never locked and fixed its root cause; **fixed** a jagged frequency axis and a sign bug.
- **Designed** an adaptive fit that counts dips first; per-line depths lifted R^2 at 0.5 A from 0.88 to 0.99.
- **Propagated** uncertainty into every field reading and the theory band.
- **Paired** with an AI assistant (Claude Code) on the code and docs; **documented** it all: README, uncertainty write-up, lab notebook, paper draft.

Python PyVISA / SCPI NumPy SciPy matplotlib Signal processing Data QC
Nonlinear fitting Uncertainty analysis

AGAINST THE OBJECTIVE

- ✓ **MET** Clean without distorting: the data-derived threshold reproduced my hand-tuned split
- ✓ **MET** Detect the resonance: a 7.48% dip, close to the roughly 8% in the literature
- ✓ **MET** Resolve the splitting: up to 8 lines, every fit $R^2 \geq 0.97$
- ⦿ **PARTLY** Match coil theory: readings ran low; one 0.61 mT offset fits the gap, and 0.3 A stays a 3.1σ outlier

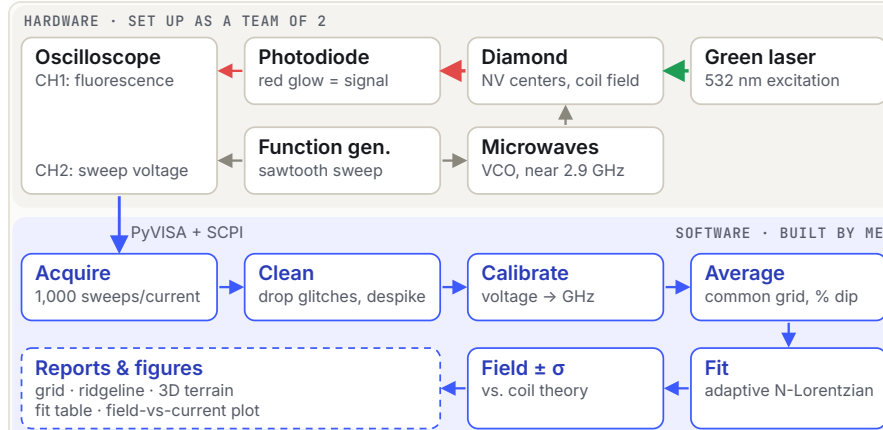


Fig. 2. The system. Gray: optics and instruments we assembled as a team. Blue: the software I built (7 Python scripts, about 2,300 lines).

CONSTRAINTS

Team: 2 people; I built all the software · **Timeline:** 7 lab sessions, analysis done by Sep 22 · **Context:** Advanced undergraduate physics lab course, fall 2026 · **Starting point:** A LabVIEW design and two- and four-dip fit models (one shared depth) from the lab handout · **Data volume:** Captures up to several GB, past GitHub's 100 MB file limit

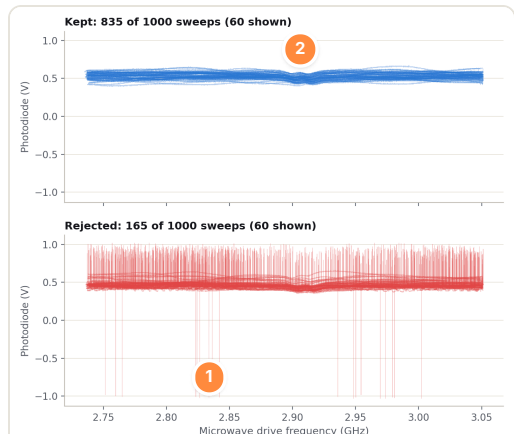


Fig. 3. Process: automated QC on the real 0 A capture.

- 1 Rejected sweeps carry single-sample spikes to ± 1 V.
- 2 Kept sweeps hint at the dip; the threshold (40.2 mV) comes from the capture's own jump statistics.

WHAT I LEARNED Build tools that make problems visible. QC plots and a read-only instrument check found most fixes; full coil geometry cut the gap from about 30σ to 3 to 13σ .

Measuring why people like mosaic art: a benchmarked 982-image feature pipeline

I wrote the MATLAB feature extractor for a WashU study of why people like mosaics: **982 mosaics** into **172 features** each. Benchmarked against a published study, it reproduces the original's average color and brightness measures (9 of 18 global features at $|r| \geq 0.7$), not yet its hue-count, blur or segment-shape features. On 70 pilot images, the team ranked features against ratings with my table.

OBJECTIVE A 2021 study predicted people's ratings of paintings from simple and complex visual features. Does that hold for mosaics? My piece: turn every image into numbers a regression can use, benchmarked against the published baseline.

982

Mosaic images in the feature table (979 extracted cleanly)

172

Features per image

23/83

Published-study features my statistics match (30 re-running its code)

70 × 5

Pilot images × ratings each

One color channel becomes 18 numbers

Lightness of all 65,536 pixels in one mosaic, after my 256 × 256 resize

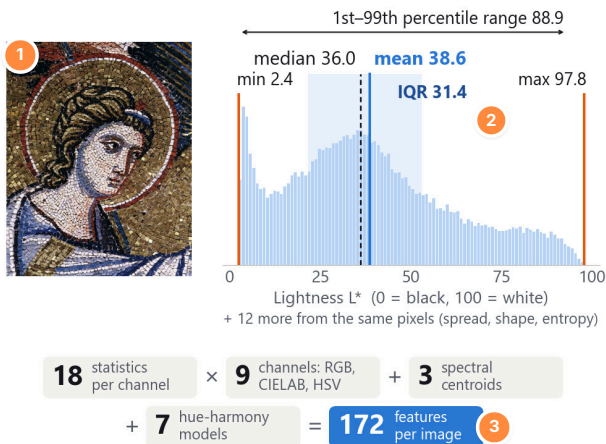


Fig. 1. What my extractor computes, for one image and one channel. Values are this image's row in my feature table.

- Input: Pietro Cavallini, *Annunciation* (detail), 1296 to 1300, public domain (photo: Web Gallery of Art), resized to 256 × 256.
- Six of the 18 statistics, drawn on the lightness histogram.
- Repeated on 9 color channels, plus frequency and color-harmony features.

MY CONTRIBUTION

Undergraduate researcher; image feature extraction and benchmarking

- Revived** the original study's MATLAB code on Windows: rebuilt its C++ binaries and checked outputs against stored values.
- Engineered** a new extractor: 18 statistics on 9 color channels plus FFT spectral centroids, reusing the original's hue-harmony models.
- Benchmarked** it against the original 83 features: 23 matched on my first pass, 30 with its hue and region code ported.
- Diagnosed** randomness in the hue-model choice (2 of 100 repeat runs changed) and in graph-cut segments.
- Evaluated** graph cuts and SAM2 for tile-level features; prototyped HOG features for tile flow.
- Delivered** the feature table for our pilot analysis; co-authored the symposium poster and abstract.

MATLAB Image processing Feature engineering Data validation Benchmarking
Correlation analysis LASSO regression Segmentation FFT Reproducibility

AGAINST THE OBJECTIVE

- ✓ **REPRODUCED** Average color and brightness: 9 of 18 global features at $|r| \geq 0.7$
- **NOT YET** The original's hue-count, blur and segment-shape features
- **PRELIMINARY** Pilot (70 images) ranks candidate features; no model validated yet

Original features matched ($|r| \geq 0.7$): 30 of 83

Benchmark on the original study's 826 paintings

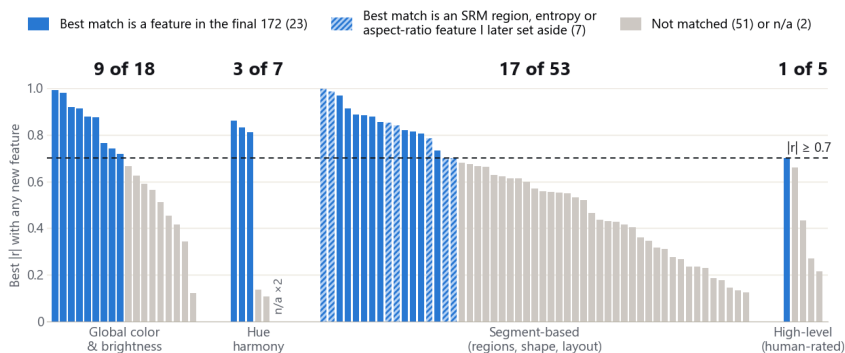
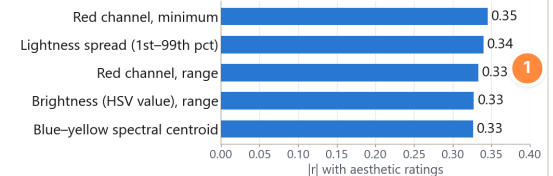


Fig. 2. Benchmark: I ran the published study's 826 paintings through my extractor and, for each of its 83 features, kept the best $|r|$ with any of mine. Hatched: region features I later set aside. Hue-model and hatched matches re-run the original study's own code, so they check my port rather than add new measurements.

Strongest single-feature correlations



Largest LASSO regression weights

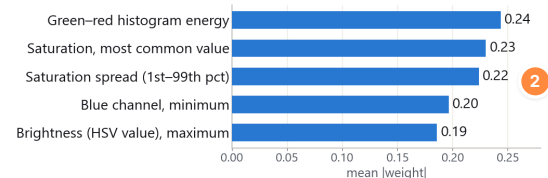


Fig. 3. Results: preliminary pilot (70 mosaics, 5 ratings each), as reported on our poster.

- All correlations are modest ($|r|$ 0.33 to 0.35); with 172 features and 70 images, this could be chance.
- LASSO ranks a different set of simple statistics.

CONSTRAINTS Team: 9 on the roster, led by Prof. Ralf Wessel; I built low-level features with Prady Sepulveda · **Ratings cost:** ~\$10 per participant-hour on Prolific (team estimate); pilot: 350 ratings · **Tooling:** Legacy MATLAB/C++ code built for macOS, rebuilt for Windows; no labeled data for deep models

WHAT I LEARNED Validation before modeling. Rebuilding legacy code, matching it feature by feature and chasing randomness is how I know which parts of the table to trust.